

Analysing and Summarising Quantitative Data

As a project manager, you may be faced with quantitative data that on its own, doesn't tell you very much; you will need to manipulate it into a different format so that it becomes more useful. This information sheet provides an overview of some key techniques which you can use to turn quantitative data into information.

Grouping Data

One way to extract meaning from data is by grouping it.

Let's look at an example. A marketing executive is working on a project to develop a new sales strategy to reward customer loyalty. She starts by looking at a spreadsheet of past orders which includes customer names, addresses, items ordered, and prices. These individual fields have limited value, so she organises it to show the highest paying customers. This reveals that the biggest orders are placed by women aged 40-50, living in the South East of England. The spreadsheet is now more revealing as a result of the grouping than in its raw form. These customers can now be targeted with special offers.

Splitting Data by Frequency

Another common approach to grouping is to split the data according to frequency - that is, how often an event occurs. It can provide you with a way to compare variations across different fields.

For example, a manager in an insurance firm is undertaking a project to examine variations in the number of claims that have been handled by departmental members over the last month. She sets up a frequency table. On the left, she enters the claims handled over the period, grouped by level. On the right, she logs the number of people who handled claims at that level. She might then look more closely at the individuals who handled the highest volume of claims for ideas on how to improve output elsewhere.

However, it's possible that the frequency table won't show up an immediately discernible pattern to establish the cause of any variations. Additionally, those with higher output may not have not been as thorough with their handling of claims. While further investigation will be needed, identifying such variations is a useful starting point.

Summary Statistics

Grouping data is *one* way to derive meaning from a set of figures. With quantitative data, we can also apply statistical techniques to draw meaning. The most common methods produce what are known as summary statistics. These include calculations such as average, range and standard deviation. They provide a snapshot - or summary - of your quantitative data.

- **The Mean Average**

Among the most common summary statistics is the **mean average**, sometimes known simply as the average or the mean. The mean average can transform a long list of numbers into a single representative figure.

For example, an HR assistant is working on a project looking to the level of departmental sickness. While the number of sick days varies across departmental team members, he calculates the mean average as 6.5 days. He can then factor this into resourcing decisions.

However, the mean average can sometimes oversimplify the data. It can obscure the range in a set of figures - this is the distance between the highest and lowest figures in a dataset. Though most team members take a modest number of days off, a handful may take two weeks or more, and another small group may take no sick days at all.

The figures at either end of this range are known as 'outliers.' They are absorbed into the average and affect its size. The more extreme the outlier, the greater the effect on the mean figure, and the less representative the average becomes. In other words, when the range is very wide, the mean average doesn't reflect reality. In this case, he could use another method to accurately represent the data - the **median**.

- **The Median**

The median is the middle figure in a data set - people often remember this because of the similarity to the word medium.

To calculate the median, he ranks the sick day figures from low to high and then selects the figure in the exact middle of the list. Of course this works only if there is an odd number of figures. If the number is even, the two central figures are added together and divided by two to provide the median.

Using the median minimises the influence of outliers - in this case the employees who take a very large or very small number of days off.

- **The Mode**

The third option he can use is the **mode**. The mode also limits distortion - it identifies the most frequently occurring number in the data set. So, if the highest number of employees take 3 sick days per year, this would be the 'mode' of sick time in the organisation.

- **Standard Deviation**

There is also a way to see how representative the mean average is of the spread, or, distribution, of any figures in a range. This is known as the **standard deviation**. It's a calculation that shows how tightly the individual figures within a range cluster around the mean. While the range provides an idea of what's going on at the extremes, the standard deviation reveals what's happening across the

distribution¹. This helps us to establish how meaningful the average is as a reflection of the data set as a whole.

For example, if you heard that graduates from a certain university go on to average starting salaries of £50,000 a year, you might be keen to apply. However, the average may be skewed by a few large outlying salaries. If you perform a standard deviation, you might find that most graduates don't earn nearly that much.

A low standard deviation figure indicates that the overall data values are close to the mean average - that is to say, they don't deviate much from the mean. A high standard deviation figure indicates that the data values have a far wider spread.

However, sometimes we'll want to see how one particular variable might affect another. There are two factors used in this analysis - correlation and causation.

Correlation and Causation

While summary statistics can provide useful insights into trends, they can't identify relationships between particular variables.

A **variable** is any characteristic that can be measured or counted. Sales figures, inches of snowfall, and marriage rates are all variables. Tracking relationships between them is one way for project managers to understand the forces at play in their decisions. The first step in analysing two variables is to look for a **correlation**.

- **Correlation**

Let's say the Assistant Buyer for an outdoors clothing company is undertaking a project to find out whether snowy weather influenced the number of woolly hats the company sold in the last two years. She can do this by looking for patterns in the data. The data values of two separate variables - sales on the one hand, and snowfall on the other - can be plotted against each other on a graph. The respective levels of each variable are represented as dots.

If there was no relationship between snow and sales, we would see no pattern among the dots. If a positive correlation is found, it would show the sales increasing as snowfall increases. We can draw a line between the dots, with an equal number on each side - the closer the dots are to the line, the stronger the correlation. However, if sales decrease as snowfall goes up, this would be a negative correlation - one variable is increasing, while the other decreases.

Spreadsheet software can calculate a figure to illustrate the strength of the correlation between the variables. This figure is known as the correlation co-efficient, or simply 'r'.

- If your spreadsheet shows an 'r' figure of zero, there is no relationship whatsoever between the two variables you have plotted.

¹ Payne, E. & Whittaker, L. (2006) *Developing Essential Study Skills* (Harlow: Pearson)

- A figure of +1 indicates a perfect positive correlation - this means an increase in one variable is matched by exactly the same increase in the other.
- A figure of -1 indicates a perfect negative correlation. This is where an increase in one variable sees a perfectly corresponding *decrease* in the other.

Between these two extremes, the strength of the correlation gradually increases and decreases. “R” figures of plus or minus 1 are rare, but the closer to 1 the figure is, the stronger the correlation.

However, the project manager may be tempted to think that because the snowfall strongly correlates with higher sales, that one is causing the other, but correlation and causation are two separate concepts.

- **Causation**

For example, it’s logical to assume that a higher number of woolly hats will be sold when snowfall is higher. But it may have more to do with another variable, such as marketing activity, or an external influence, such as celebrity hat-wearing trends. Equally, it could be a total coincidence.

In some correlations, it appears the variables are linked, but it’s unclear which is having the effect. In others, a third variable may be responsible for the increases to both observed variables. A commonly cited example is the correlation between ice cream sales and homicide rates; when one increases, so does the other. However, it’s likely that the driver in both increases is a third factor - hot summer weather.

- **A ‘confounding factor’**

This is known as a ‘confounding factor’ - a third variable that distorts the relationship between the two variables being studied. For example, some time ago a large medical study indicated that women taking hormone replacement therapy were less likely to suffer from heart disease. Based on this association, hormone therapy was judged as having a protective effect on the heart.

However, there was a confounding factor. Wealthier women tend to have better health - including healthier hearts - and these were the women more likely to seek hormone replacement from their doctors. The protective factor in this case wasn’t the hormones - it was money.

Looking for correlations in your data can provide an excellent starting point for uncovering causal relationships. Just keep in mind that to prove causation you’ll need to undertake further analysis.